

Deep research prompt

对框架的理解摘要：

“框架”文件构建了一个用于分析/比较网格生成算法的统一视角，核心思想是将算法分解为一系列明确的模块和规则，每个模块承担不同的职责。框架结构及规则如下：

模块名称	核心内容	作用
Input (data, problem definition)	- <i>data</i> : 算法的原始输入数据。 - <i>problem definition</i> : 对网格生成任务的正式定义（如需要生成 n 边形网格、原始数据的维数等）。	明确需要解决的问题是什么以及有哪些必备的初始数据。
Global Rule	对所有网格生成问题都适用的几何和拓扑规则，不含算法特有的规则。	为算法设定不可违反的通用约束，保证生成网格的有效性。
Problem reduction	根据 <i>problem definition</i> 和 Global Rule，将原始问题映射为一个已经解决或更简单的问题；如果定义与全局规则冲突，需判断问题是否无解。	简化或重新定义问题，使之适合特定算法处理。
Input (data, problem definition ')	- <i>data</i> : 仍然是最初的输入数据。 - <i>problem definition '</i> : 经过化简后的问题定义。	准备算法真正使用的输入，即经过“问题化简”后的版本。
Local Rule	特定算法的专属规则或启发式规则；并非所有算法都需要。	用于体现某个算法的独特思路，解决全局规则未涵盖的细节问题。
Engine	接收 <i>data</i> 、 <i>problem definition '</i> 、 <i>Global Rule</i> 和 <i>Local Rule</i> 作为输入，包含： - <i>objective</i> : 算法努力达到的目标，在达成之前会持续执行方法。 - <i>method</i> : 为完成目标而采取的具体求解步骤，必须遵循全局与局部规则。	算法的执行核心，用来完成网格生成任务。

数据库结构和字段规范

分类	字段名称	说明
基本信息	Algorithem name、Detail Name	分别给出算法的简短名称与详细名称或论文标题。
维度与类型	Dimension、Element Type	以数组形式标注问题维度（如 2D / 3D ）以及网格元素类型（如 Quad 、 Tri 等）。
属性评价	Era、Computational Cost、Practical Maturity、Method Paradigm、Resource Type、Theoretical Guarantees	用简短词组或列表描述算法所属年代、计算代价、成熟度、方法范式（如 Learning ）、资源类型（如 论文 ）、理论保证等。
适用几何	Geometric Complexity	数组，列举可处理的几何形状难点（如 尖角 、 非凸 、 带孔 ）。
规则与问题定义	Global Rule、Problem Reduction	用自然语言条目描述全局几何拓扑规则、将网格生成任务简化为形式化问题的步骤。
输入	Input	说明算法所需的原始边界、密度参数及超参数等，必要时分行列出。
局部规则	Local Rule	阐述特定算法的局部选择策略、观测定义、动作范围与奖励项。
引擎细节	Engine.method、Engine.objective	用句子说明执行算法的核心方法与目标。例如 Engine.method 描述具体运算流程或学习策略， Engine.objective 描述优化目标或累积奖励。
输出	Output	叙述生成网格的性质，如元素质量、对齐性和密度分布等。
依赖与参考	Dependencies、References	Dependencies 列出必需的边界条件、几何运算及工具； References 提供文献信息或外部链接，常用 BibTeX 格式。
性能与优势	Core Advantages、Computational Cost	解释算法的核心优点及成本。
建议与局限	Complementary Suggestions、Main Limitations、Failure Modes、Framework Conformance, Typical Deviations	提供与其他算法的配合建议，列出主要局限、失败模式以及常见偏差或需要注意的超参数敏感性等。

算法列表：

Advancing Front Method

- Detail name: Advancing Front Mesh Generation Method
- Reference:

```
@article{Lo1985,  
  author = {Siu-Hang Lo},  
  title  = {A New Mesh Generation Scheme for Arbitrary Planar Domains},  
  journal = {International Journal for Numerical Methods in Engineering},  
  year   = {1985},  
  volume = {21},  
  number = {8},  
  pages  = {1403--1426},  
  doi    = {10.1002/nme.1620210805}  
}
```

Bubble Mesh

- Detail name: Bubble Mesh: Sphere-Packing-Based Triangular Mesh Generation
- Reference:

```
@article{Shimada1998,  
  author = {Kenji Shimada},  
  title  = {Bubble Mesh: Automated Triangular Meshing of Non-Manifold Geometry by Sphere Packing},  
  journal = {Computer-Aided Geometric Design},  
  year   = {1998},  
  volume = {15},  
  number = {3},  
  pages  = {199--222},  
  doi    = {10.1016/S0167-8396(97)00033-X}
```

}

Chew's Refinement

- Detail name: Chew's Delaunay Refinement Algorithm for Quality Triangulation
- Reference:

@article{Chew1989,

author = {L. Paul Chew},

title = {Constrained Delaunay Triangulations},

journal = {Algorithmica},

year = {1989},

volume = {4},

pages = {97--108},

doi = {10.1007/BF01553881}

}

@inproceedings{ChewSecondAlgorithm,

author = {L. Paul Chew},

title = {Guaranteed-Quality Mesh Generation for Curved Surfaces},

booktitle = {Proceedings of the Ninth Annual Symposium on Computational Geometry},

year = {1993},

pages = {274--280}

}

@techreport{Shewchuk1997},

author = {Jonathan Richard Shewchuk},

title = {Delaunay Refinement Algorithms for Triangular Mesh Generation},

institution = {Carnegie Mellon University},

year = {1997}

}

CVT / Lloyd Optimization

- Detail name: Centroidal Voronoi Tessellation (CVT) / Lloyd Optimization
- Reference:

```
@article{Du1999,  
  author = {Qiang Du and Vance Faber and Max Gunzburger},  
  title  = {Centroidal Voronoi Tessellations: Applications and Algorithms},  
  journal = {SIAM Review},  
  year   = {1999},  
  volume = {41},  
  number = {4},  
  pages  = {637--676},  
  doi    = {10.1137/S0036144599352836}  
}
```

Delaunay Triangulation (Bowyer–Watson)

- Detail name: Bowyer–Watson Incremental Delaunay Triangulation Algorithm
- Reference:

```
@article{Bowyer1981,  
  author = {Adrian Bowyer},  
  title  = {Computing Dirichlet Tessellations},  
  journal = {The Computer Journal},  
  year   = {1981},  
  volume = {24},  
  number = {2},  
  pages  = {162--166},  
  doi    = {10.1093/comjnl/24.2.162}  
}
```

DistMesh

- Detail name: DistMesh: Distance-Field-Based Spring Relaxation Mesh Generator
- Reference:

```
@article{Persson2004,
  author = {Per-Olof Persson and Gilbert Strang},
  title = {A Simple Mesh Generator in MATLAB},
  journal = {SIAM Review},
  year = {2004},
  volume = {46},
  number = {2},
  pages = {329--345},
  doi = {10.1137/S0036144503429121}
}
```

FreeMesh-RL

- Detail name: FreeMesh-RL: SAC-Based Reinforcement Learning for Automatic Quadrilateral Mesh Generation
- Reference:

```
@article{Pan2022FreeMeshRL,
  author = {Jie Pan and Jingwei Huang and Gengdong Cheng and Yong Zeng},
  title = {Reinforcement learning for automatic quadrilateral mesh generation: a soft actor-critic approach},
  journal = {Neural Networks},
  year = {2022},
  note = {Preprint submitted to Neural Networks, arXiv:2203.11203}
}
```

链接:

<https://arxiv.org/abs/2203.11203%22%EF%BC%88%E5%8F%AF%E5%9C%A8>

Instant Meshes

- Detail name: Instant Meshes: Instant Field-Aligned Quadrilateral Meshing
- Reference:

```
@inproceedings{Jakob2015,
  author = {Wenzel Jakob and Marco Tarini and Daniele Panozzo and Olga
Sorkine-Hornung},
  title = {Instant Field-Aligned Meshes},
  booktitle = {ACM Transactions on Graphics (TOG), Proceedings of
SIGGRAPH},
  year = {2015},
  volume = {34},
  number = {6},
  pages = {189:1--15},
  doi = {10.1145/2816795.2818078}
}
```

MIQ (Mixed-Integer Quadrangulation)

- Detail name: Mixed-Integer Quadrangulation (MIQ)
- Reference:

```
@article{Bommes2009,
  author = {David Bommes and Henrik Zimmer and Leif Kobbelt},
  title = {Mixed-Integer Quadrangulation},
  journal = {ACM Transactions on Graphics},
  year = {2009},
  volume = {28},
  number = {3},
  pages = {77:1--10},
  doi = {10.1145/1531326.1531383}
}
```

Paving

- Detail name: Paving: All-Quadrilateral Advancing-Front Mesh Generation
- Reference:

```
@article{Blacker1991,
  author = {Ted D. Blacker and Michael B. Stephenson},
  title = {Paving: A New Approach to Automated Quadrilateral Mesh
  Generation},
  journal = {International Journal for Numerical Methods in Engineering},
  year = {1991},
  volume = {32},
  number = {4},
  pages = {811--847},
  doi = {10.1002/NME.1620320410}
}
```

Plastering

- Detail name: Plastering: All-Hexahedral Advancing-Front Filling Method
- Reference:

```
@article{Blacker1993,
  author = {Ted D. Blacker and Ray J. Meyers},
  title = {Seams and Wedges in Plastering: A 3-D Hexahedral Mesh
  Generation Algorithm},
  journal = {Engineering with Computers},
  year = {1993},
  volume = {9},
  pages = {83--93},
  doi = {10.1007/BF01199047}
}
```

Q-Morph

- Detail name: Q-Morph: Triangle-to-Quadrilateral Conversion Method

- Reference:

```
@article{Owen1999,
  author = {Steven J. Owen and Mark L. Staten and Scott A. Canann and
    Sanjay Saigal},
  title = {Q-Morph: An Indirect Approach to Advancing Front Quad Meshing},
  journal = {International Journal for Numerical Methods in Engineering},
  year = {1999},
  volume = {44},
  number = {9},
  pages = {1317--1340},
  doi = {10.1002/(SICI)1097-0207(19990330)44:9<1317::AID-NME486>
}
```

Quadtree / Octree Subdivision

- Detail name: Quadtree/Octree-Based Mesh Subdivision Method
- Reference:

```
@article{Yerry1983,
  author = {Mark A. Yerry and Mark S. Shephard},
  title = {A Modified Quadtree Approach to Finite Element Mesh Generation},
  journal = {IEEE Computer Graphics and Applications},
  year = {1983},
  volume = {3},
  number = {1},
  pages = {39--46},
  doi = {10.1109/MCG.1983.262997}
}
```

Ruppert's Refinement

- Detail name: Ruppert's Delaunay Refinement Algorithm for Quality 2D Mesh Generation

- Reference:

```
@article{Ruppert1995,
  author = {Jim Ruppert},
  title = {A Delaunay Refinement Algorithm for Quality 2-Dimensional Mesh
  Generation},
  journal = {Journal of Algorithms},
  year = {1995},
  volume = {18},
  number = {3},
  pages = {548--585},
  doi = {10.1006/jagm.1995.1021}
}
```

Structured Mapping / PDE-Based Mesh Generation

- Detail name: Structured Mapping / PDE-Based Mesh Generation Method
- Reference:

```
@article{Winslow1966},
  author = {A. M. Winslow},
  title = {Numerical Solution of the Quasilinear Poisson Equation in a
  Nonuniform Triangle Mesh},
  journal = {Journal of Computational Physics},
  year = {1966},
  volume = {1},
  number = {2},
  pages = {149--172},
  doi = {10.1016/0021-9991(66)90007-7}
}
```

数据行必备属性 (Required attributes)

- 基本标识
 - Algorithm name（必填，标题列）
 - Detail Name（论文标题或方法全名）
- 引用与来源
 - References（BibTeX）
 - Publication Date（年份或日期，可为单日）
 - Resource Type（论文、书籍、工具实现等）
- 定位与分类
 - Era（年代）
 - Method Paradigm（方法范式，例：Delaunay、Optimization、Learning）
- 维度与网元
 - Dimension（2D / 3D / Surface / Volume，可多选）
 - Element Type（Tri / Quad / Tet / Hex / Mixed，可多选）
- 规则与机制
 - Global Rule（全局约束）
 - Problem Reduction（问题化简）
 - Local Rule（局部规则，若适用）
 - Engine.method（核心方法）
 - Engine.objective（优化/求解目标）
- 输出与质量
 - Output（输出网格性质）
 - Quality Metrics（最小角阈值、各向同性、方向对齐等，可多选）
 - Theoretical Guarantees（终止性、最小角下界等，可多选）
- 实践信息
 - Computational Cost（低/中/高）
 - Practical Maturity（研究探索/工程可用/经典成熟）
- 依赖与补充

- Dependencies（必需数据、工具链、前置算法）
- Complementary Suggestions（与其他方法协同建议）
- Main Limitations（主要局限）
- Failure Modes（失败模式）
- Framework Conformance（描述算法与框架之间的符合程度，如果不符合则在Typical Deviations种给出解释）
- Typical Deviations（常见偏差：说明该算法与框架产生偏差的原因）